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年級:

科目:

練習題 19.4

$$1. (2) \nabla \times \vec{F} = \left(\frac{\partial R}{\partial y} - \frac{\partial Q}{\partial z}, \frac{\partial P}{\partial z} - \frac{\partial R}{\partial x}, \frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) = (0, 0, 0)$$

所以 $\vec{F}$ 是無旋場 (在 $\mathbb{R}^3 \setminus \{(0,0,0)\}$ 中) $\Rightarrow \vec{F}$ 是有勢場

$$\begin{aligned} \varphi(x, y, z) &= \int_{(0,0,0)}^{(x,y,z)} \vec{F} \cdot d\vec{p} = \int_{(0,0,0)}^{(x,0,0)} \vec{F} \cdot d\vec{p} + \int_{(x,0,0)}^{(x,y,0)} \vec{F} \cdot d\vec{p} + \int_{(x,y,0)}^{(x,y,z)} \vec{F} \cdot d\vec{p} \\ &= \int_0^x \frac{1}{t} dt + \int_0^y \frac{1}{t} dt + \int_{\sqrt{(x+y)^2+z^2}}^{\sqrt{(x+y)^2+z^2}} \frac{1}{t^2} \cdot \frac{1}{2} \cdot d(t^2) = \ln \left| \frac{x+y}{1} \right| + \frac{1}{2} \ln \left(\frac{(x+y)^2+z^2}{(x+y)^2} \right) \\ &= \frac{1}{2} \ln [(x+y)^2+z^2] \end{aligned}$$

$$2. \int_{(1,2,3)}^{(0,1,1)} yz dx + xz dy + xy dz = \int_{(1,2,3)}^{(0,2,3)} + \int_{(0,2,3)}^{(0,1,3)} + \int_{(0,1,3)}^{(0,1,1)} = \int_1^0 6 dx + \int_2^1 0 dy + \int_3^1 0 dz = -6$$

$$3. \nabla \times (f, g, h) = (0, 0, 0). \text{ 所以 } \exists u, \text{ 使得 } \nabla u = (f, g, h)$$

$$\text{所以 } \int_{(x_1,y_1,z_1)}^{(x_2,y_2,z_2)} \vec{F} \cdot d\vec{p} = \int_{(x_1,y_1,z_1)}^{(x_2,y_2,z_1)} + \int_{(x_2,y_2,z_1)}^{(x_2,y_2,z_2)} = \int_{x_1}^{x_2} f(x) dx + \int_{y_1}^{y_2} g(y) dy + \int_{z_1}^{z_2} h(z) dz$$

$$4. (1) \omega = f(x+y+z)(dx+dy+dz)$$

$$d\omega = \frac{df}{dt} \cdot (dx+dy+dz) \wedge (dx+dy+dz) = 0 \Leftrightarrow f(x,y,z) \cdot (1,1,1) \text{ 是有勢場}$$

$$\int_{(x_1,y_1,z_1)}^{(x_2,y_2,z_2)} \omega = \int_{(x_1,y_1,z_1)}^{(x_2,y_2,z_1)} + \int_{(x_2,y_2,z_1)}^{(x_2,y_2,z_2)} = \int_{x_1+y_1+z_1}^{x_2+y_2+z_2} f(t) dt + \int_{x_1+y_1+z_1}^{x_2+y_2+z_2} f(t) dt + \int_{x_1+y_1+z_1}^{x_2+y_2+z_2} f(t) dt = \int_{x_1+y_1+z_1}^{x_2+y_2+z_2} f(t) dt$$

$$5. \vec{F} = k(x, y) \quad (k < 0). \quad \text{令 } x = a \cos t, y = b \sin t, t \in [0, \frac{\pi}{2}]$$

$$\int_{(a,0)}^{(0,b)} \vec{F} \cdot d\vec{p} = \int_{(a,0)}^{(0,b)} kx dx + ky dy = \int_{(a,0)}^{(0,0)} kx dx + \int_{(0,0)}^{(0,b)} ky dy = \frac{k}{2} (0^2 - a^2 + b^2 - 0^2) = \frac{k}{2} (b^2 - a^2)$$

$$6. (3) \frac{\partial(2x+y)}{\partial x} = 2 = \frac{\partial(x+2y)}{\partial y} \Rightarrow \exists u, \nabla u = (x+2y, 2x+y)$$

$$u = \int_{(0,0)}^{(x,y)} (x+2y, 2x+y) \cdot (dx, dy) = \int_{(0,0)}^{(x,0)} t dt + \int_{(x,0)}^{(x,y)} (2x+y) dy = \frac{1}{2}x^2 + \frac{1}{2}[(2x+y)^2 - (2x)^2] = C.$$

$$\Leftrightarrow x^2 + 4xy + y^2 = C.$$

$$(4) \frac{\partial[-(x+\sin^2 y)]}{\partial x} = -1 = \frac{\partial(x^2 y)}{\partial y} \Rightarrow (x^2 y, -x - \sin^2 y) \text{ 是有势场}$$

$$\begin{aligned} \varphi(x, y) &= \int_{(0,0)}^{(x,y)} = \int_{(0,0)}^{(x,0)} + \int_{(x,0)}^{(x,y)} = \int_0^x t^2 dt - \int_{(x,0)}^{(x,y)} (x + \sin^2 t) dt = \frac{1}{3}x^3 - xy + \int_0^y \frac{\cos 2t - 1}{2} dt \\ &= \frac{1}{3}x^3 - xy + \frac{1}{2} \left(\frac{1}{2} \sin 2y - y \right) = C. \end{aligned}$$

$$(5) \frac{\partial(xe^y - 2y)}{\partial x} = e^y = \frac{\partial(e^y)}{\partial y} \Rightarrow (e^y, xe^y - 2y) \text{ 是有势场}$$

$$\begin{aligned} \varphi(x, y) &= \int_{(0,0)}^{(x,y)} = \int_{(0,0)}^{(x,0)} + \int_{(x,0)}^{(x,y)} = \int_0^x e^0 dx + \int_0^y (xe^y - 2y) dy = x + x(e^y - 1) - y^2 = C. \\ \Leftrightarrow xe^y - y^2 &= C. \end{aligned}$$

練習題 20.1

$$1. (1) \lim_{a \rightarrow 0} \int_{-1}^1 \sqrt{x^2 + a^2} dx = \int_{-1}^1 \lim_{a \rightarrow 0} \sqrt{x^2 + a^2} dx = \int_{-1}^1 |x| dx = 2 \int_0^1 x dx = x^2 \Big|_0^1 = 1$$

$$(2) \lim_{t \rightarrow 0} \int_0^2 x^2 \cos tx dx = \int_0^2 x^2 dx = \frac{1}{3} x^3 \Big|_0^2 = \frac{8}{3}$$

$$2. F'(u) = u f(u) + \int_0^u f(x) dx + u f(u) = \int_0^u f(x) dx + 2u f(u)$$

$$(3) F'(u) = f(u) + 2(f(u) + u f'(u)) = 3f(u) + 2u f'(u)$$

$$3. \lim_{h \rightarrow 0} \frac{1}{h} \int_a^x (f(t+h) - f(t)) dt = \int_a^x \lim_{h \rightarrow 0} \frac{1}{h} (f(t+h) - f(t)) dt = \int_a^x f'(t) dt = f(x) - f(a)$$

4.

$$\begin{aligned} 5. \quad \frac{\partial u}{\partial x} &= \frac{1}{2} \left(\frac{\partial \psi}{\partial x}(x-at) - \frac{\partial \psi}{\partial x}(x+at) \right) + \frac{1}{2a} \int_{-at}^{at} \frac{d\psi}{dx}(x+s) ds \\ &= \frac{1}{2} \left(\frac{\partial \psi}{\partial x}(x-at) - \frac{\partial \psi}{\partial x}(x+at) \right) + \frac{1}{2a} [\psi(x+at) - \psi(x-at)] \\ \frac{\partial^2 u}{\partial x^2} &= \frac{1}{2} \left[\frac{\partial^2 \psi}{\partial x^2}(x-at) - \frac{\partial^2 \psi}{\partial x^2}(x+at) \right] + \frac{1}{2a} \left[\frac{\partial \psi}{\partial x}(x+at) - \frac{\partial \psi}{\partial x}(x-at) \right] \\ \frac{\partial^2 u}{\partial t^2} &= \frac{1}{2} \left[(-a)^2 \frac{\partial^2 \psi}{\partial t^2}(x-at) - a^2 \frac{\partial^2 \psi}{\partial t^2}(x+at) \right] + \frac{1}{2a} \left(\frac{\partial \psi}{\partial t}(x+at) - \frac{\partial \psi}{\partial t}(x-at) \right) \\ &= a^2 \frac{\partial^2 u}{\partial x^2} \end{aligned}$$

$$6. \frac{\partial u}{\partial y} = - \int_0^a \frac{y f(t) dt}{[(x-t)^2 + y^2 + z^2]^{\frac{3}{2}}}, \quad \frac{\partial^2 u}{\partial y^2} = - \int_0^a f(t) \cdot \frac{3}{2} \sqrt{(x-t)^2 + y^2 + z^2} -$$

$$\frac{\partial^2 u}{\partial y^2} = - \int_0^1 \frac{f(t) [(x-t)^2 + y^2 + z^2]^{\frac{3}{2}} - \frac{2}{3} f(t) y^2 \sqrt{(x-t)^2 + y^2 + z^2}}{[(x-t)^2 + y^2 + z^2]^{\frac{3}{2}}} dt$$

$$\frac{\partial^2 u}{\partial x^2} = \frac{\partial^2 u}{\partial (x-y)^2}, \quad \frac{\partial^2 u}{\partial x^2} \quad \text{由} \quad \frac{\partial^2 u}{\partial (x-y)^2}, \quad \frac{\partial^2 u}{\partial y^2}, \quad \frac{\partial^2 u}{\partial z^2} \quad \text{輪換對稱。}$$

所以 $\Delta u = - \int_0^a \frac{3f(t) [(x-t)^2 + y^2 + z^2]^{\frac{3}{2}} - 3f(t) \cdot ((x-t)^2 + y^2 + z^2) \sqrt{(x-t)^2 + y^2 + z^2}}{[(x-t)^2 + y^2 + z^2]^3} dt = - \int_0^a 0 dt = 0$

7. $\varphi'(u) = \int_a^b f(x) \cdot \frac{x-u}{|x-u|} dx$. $\frac{x}{|x|}$, $u \notin (a, b)$, 则 $\varphi'(u) = \int_a^b f(x) dx$ 或 $-\int_a^b f(x) dx$

此时 $\varphi'(u) = 0$.

若 $u \in [a, b]$, 則 $\varphi'(u) = -\int_a^u f(x) dx - \int_b^u f(x) dx \Rightarrow \varphi'(u) = -f(u) - f(u) = -2f(u)$

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$$y = a \tan x$$

$$dy = a(\tan^2 x + 1) dx = a\left(\frac{y^2}{a^2} + 1\right) dx = \frac{y^2 + a^2}{a} dx$$

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$$\begin{aligned} 2. (3) \int_0^{\frac{\pi}{2}} \frac{\arctan(a \tan x)}{\tan x} dx, \quad \text{令 } y = a \tan x \Rightarrow dy &= a(\tan^2 x + 1) dx = \frac{1}{a}(y^2 + a^2) dx \\ &= \int_0^{+\infty} \frac{\arctan y}{\frac{y}{a}} \cdot \frac{a}{y^2 + a^2} dy = a^2 \int_0^{+\infty} \frac{\arctan y}{y^2} \cdot \frac{y dy}{y^2 + a^2} \\ &= \frac{a^2}{2} \int_0^{+\infty} \frac{\arctan y}{y^2(y^2 + a^2)} \cdot d(y^2) = \frac{1}{2} \int_0^{+\infty} \left(\frac{\arctan y}{y^2} - \frac{\arctan y}{y^2 + a^2} \right) dy^2 \\ &= \frac{1}{2} \left[\arctan y \ln\left(\frac{y^2}{y^2 + a^2}\right) \right]_0^{+\infty} - \int_0^{+\infty} \frac{y^2}{(y^2 + a^2)^2} \cdot \frac{1}{1 + y^2} dy \\ &= \frac{1}{2} \int_0^{+\infty} \ln\left(1 + \frac{a^2}{y^2}\right) \cdot \frac{1}{1 + y^2} dy \end{aligned}$$

$$(3) \int_0^{\frac{\pi}{2}} \frac{\arctan(a \tan x)}{\tan x} dx$$

$$\text{解. 令 } \tan \beta = a \tan x, \quad (\tan^2 \beta + 1) d\beta = a(\tan^2 x + 1) dx = \frac{1}{a}(\tan^2 \beta + a^2) dx$$

$$\int_0^{\frac{\pi}{2}} \frac{\arctan(a \tan x)}{\tan x} dx = \int_0^{\frac{\pi}{2}} \frac{\beta}{\tan \beta} \cdot \frac{a(\tan^2 \beta + 1)}{a^2 + \tan^2 \beta} d\beta$$

$$\text{設所求積分是 } F(a) = \int_0^{\frac{\pi}{2}} \frac{\arctan(a \tan x)}{\tan x} dx$$

$$F'(a) = \int_0^{\frac{\pi}{2}} \frac{1}{\tan x} \cdot \frac{\tan x}{a^2 \tan^2 x + 1} dx = \int_0^{\frac{\pi}{2}} \frac{1}{1 + a^2 \tan^2 x} dx \quad (\text{有 } F'(a) = F'(-a)).$$

$$(\text{不妨設 } a > 0) \int_0^{+\infty} \frac{1}{1 + y^2} \cdot \frac{a}{y^2 + a^2} dy = \frac{a}{a^2 - 1} \int_0^{+\infty} \left(\frac{1}{y^2 + 1} - \frac{1}{y^2 + a^2} \right) dy$$

$$= \frac{a}{a^2 - 1} \left[\arctan y \right]_0^{+\infty} - \frac{1}{a} \arctan \frac{y}{a} \Big|_0^{+\infty} = \frac{a}{a^2 - 1} \left(\frac{\pi}{2} - \frac{1}{a} \cdot \frac{\pi}{2} \right)$$

$$= \frac{1}{a+1} \cdot \frac{\pi}{2}$$

$$\text{所以 } F'(a) = \begin{cases} \frac{\pi}{2} \cdot \frac{1}{a+1}, & a \geq 0 \\ \frac{\pi}{2} \cdot \frac{1}{1-a}, & a < 0 \end{cases} \Rightarrow F(a) = \begin{cases} \frac{\pi}{2} \ln(1+a), & a \geq 0 \\ -\frac{\pi}{2} \ln(1-a), & a < 0 \end{cases} = \frac{\pi}{2} \cdot \text{sgn}(a) \cdot \ln(1+|a|)$$

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練習題 20.2.

$$(2) \int_{-\infty}^{+\infty} \frac{x^2 \cos ux}{1+x^6} dx = 2 \int_0^{+\infty} \frac{x^2 \cos ux}{1+x^6} dx, \quad \text{而 } \left| \frac{x^2 \cos ux}{1+x^6} \right| \leq \frac{1}{x^2}, \quad \int_0^{+\infty} \frac{1}{x^2} dx =$$

$$\text{而 } \left| \frac{x^2 \cos ux}{1+x^6} \right| \leq \frac{x^2}{1+x^6}, \quad \int_0^{+\infty} \frac{x^2}{1+x^6} dx = \int_1^{+\infty} \frac{x^2}{1+x^6} dx + \int_{+\infty}^1 \frac{(\frac{1}{t})^2}{1+(\frac{1}{t})^6} d(\frac{1}{t})$$

$$= \int_1^{+\infty} \frac{x^2}{1+x^6} dx + \int_1^{+\infty} \frac{(\frac{1}{t})^4}{1+(\frac{1}{t})^6} dt = \int_1^{+\infty} \frac{1}{t^4} dt + \int_1^{+\infty} \frac{x^2}{1+x^6} dx = \int_1^{+\infty} \frac{x^2+1}{x^6+1} dx$$

$$\text{而 } \frac{x^2+1}{x^6+1} \sim \frac{1}{x^4} \quad (x \rightarrow +\infty), \quad \text{所以 } \int_0^{+\infty} \frac{x^2}{1+x^6} dx \text{ 收斂}$$

$$\text{由 Weierstrass 定理知 } \int_{-\infty}^{+\infty} \frac{x^2 \cos ux}{1+x^6} dx \text{ 一致收斂}$$

$$(3) \int_0^{+\infty} \frac{dx}{1+(x+u)^2} = \arctan(x+u) \Big|_0^{+\infty} = \frac{\pi}{2} - \arctan u$$

$$\text{解 } \int_A^{+\infty} \left| \frac{1}{1+(x+u)^2} \right| dx \leq \int_A^{+\infty} \frac{1}{1+x^2} dx = \arctan \frac{\pi}{2} - \arctan A$$

一致收斂

$$(5) \int_0^{+\infty} \sqrt{n} e^{-ux^2} dx = \frac{\sqrt{\pi}}{2}, \quad \int_A^{+\infty} \sqrt{n} e^{-ux^2} dx = \int_{\sqrt{n}A}^{+\infty} e^{-x^2} dx, \quad \text{取 } u = \frac{1}{n^2}$$

~~$$\text{取 } u_1, u_2, 0 < u_1 < u_2$$~~
~~$$\int_A^{+\infty} \sqrt{u_1} e^{-u_1 x^2} dx = \int_A^{+\infty} \sqrt{u_2} e^{-u_2 x^2} dx$$~~

$$\forall n \in \mathbb{N}^+, \quad \int_n^{+\infty} \sqrt{n} e^{-ux^2} dx = \int_1^{+\infty} e^{-x^2} dx > 0.$$

不一致收斂

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練習題 20.3

1. (1) $\forall x \in (2, +\infty), \exists \delta \in [2, x]$

$$\int_0^{+\infty} \frac{t}{2+t^x} dt = \int_0^1 \frac{t}{2+t^x} dt + \int_1^{+\infty} \frac{t}{2+t^x} dt$$

$$\left| \int_A^{+\infty} \frac{t}{2+t^x} dt \right| < \int_A^{+\infty} \frac{1}{t^{x-1}} dt \text{ 收斂}$$

所以 $\int_0^{+\infty} \frac{t}{2+t^x} dt$ 在 $[2, +\infty)$ 內一致連續 $\Rightarrow f(x)$ 在 $[2, +\infty)$ 內連續

$\Rightarrow f(x)$ 在 $(2, +\infty)$ 內連續

$$(2) \int_0^\pi \frac{\sin x}{x^\alpha(\pi-x)^\alpha} dx = \int_0^{\frac{\pi}{2}} \frac{\sin x}{x^\alpha(\pi-x)^\alpha} dx + \int_{\frac{\pi}{2}}^\pi \frac{\sin x}{x^\alpha(\pi-x)^\alpha} dx$$

$\forall \alpha \in (0, 2), \exists \delta \in (x, 2)$

$$\left| \int_0^\delta \frac{\sin x}{x^\alpha(\pi-x)^\alpha} dx \right| \leq \frac{1}{\left(\frac{\pi}{2}\right)^\alpha} \left| \int_0^\delta \frac{1}{x^{\alpha-1}} dx \right| < \int_0^\delta \frac{1}{x^{c-1}} dx = \frac{1}{2-c} \delta^{2-c} \text{ 收斂}$$

$$\left| \int_\beta^\pi \frac{\sin x}{x^\alpha(\pi-x)^\alpha} dx \right| = \left| \int_{\pi-\beta}^0 \frac{\sin(\pi-x)}{x^\alpha(\pi-x)^\alpha} (-dx) \right| = \left| \int_0^{\pi-\beta} \frac{\sin(\pi-x)}{x^\alpha(\pi-x)^\alpha} dx \right| < \frac{1}{2-c} (\pi-\beta)^{2-c} \text{ 收斂}$$

由 Weierstrass 定理知, $\varphi(\alpha)$ 在 $(0, c)$ 內一致收斂

$\Rightarrow \varphi(\alpha)$ 在 $(0, c)$ 內連續

$\Rightarrow \varphi(\alpha)$ 在 $(0, 2)$ 內連續

$$3. \text{ 設 } I_1(a) = \int_0^{+\infty} e^{-ax} \cdot \frac{\sin cx}{x} dx, \quad I_1'(a) = \int_0^{+\infty} -e^{-ax} \sin cx dx = \int_0^{+\infty} e^{-ax} \cdot \frac{1}{c} d(\cos cx)$$

$$= \frac{1}{c} \left[e^{-ax} \cos cx \right]_0^{+\infty} - \int_0^{+\infty} \cos cx \cdot (-a) e^{-ax} dx = \frac{1}{c} \left[-1 + \frac{a}{c} \int_0^{+\infty} e^{-ax} d(\sin cx) \right]$$

$$= \frac{1}{c} \left[\frac{a}{c} \left(e^{-ax} \sin cx \right)_0^{+\infty} - \int_0^{+\infty} \sin cx \cdot (-a) e^{-ax} dx - 1 \right] = \left(\frac{a}{c} \right)^2 (-I_1'(a)) - \frac{1}{c}$$

$$I_1'(a) = \frac{-\frac{1}{c}}{1 + \left(\frac{a}{c}\right)^2} = -\frac{c}{c^2 + a^2}, \quad I_1(a) = -\arctan\left(\frac{a}{c}\right) + E$$

$$I_1(+\infty) = 0 \Rightarrow E = \frac{\pi}{2}$$

$$\Rightarrow I_1(a) = \frac{\pi}{2} - \arctan\left(\frac{a}{c}\right)$$

$$\text{所以 } \int_0^{+\infty} \frac{e^{-ax} - e^{-bx}}{x} \sin x \, dx = \frac{\pi}{2} - \arctan\left(\frac{a}{c}\right) - \left[\frac{\pi}{2} - \arctan\left(\frac{b}{c}\right) \right] = \arctan\left(\frac{b}{c}\right) - \arctan\left(\frac{a}{c}\right)$$

$$4. \text{ 记 } I(\alpha) = \int_0^{+\infty} \frac{\log(\alpha^2 + x^2)}{\beta^2 + x^2} \, dx, \quad \frac{\log(\alpha^2 + x^2)}{\beta^2 + x^2} \sim \frac{2 \log x}{x^2} < \frac{2}{x^{1.5}} \Rightarrow I(\alpha) \text{ 一致收敛}$$

$$I'(\alpha) = \int_0^{+\infty} \frac{2\alpha}{(\alpha^2 + x^2)(\beta^2 + x^2)} \, dx = \frac{2\alpha}{\beta^2 - \alpha^2} \int_0^{+\infty} \left(\frac{1}{\alpha^2 + x^2} - \frac{1}{\beta^2 + x^2} \right) \, dx$$

$$= \frac{2\alpha}{\beta^2 - \alpha^2} \left[\frac{1}{\alpha} \arctan\left(\frac{x}{\alpha}\right) \Big|_0^{+\infty} - \frac{1}{\beta} \arctan\left(\frac{x}{\beta}\right) \Big|_0^{+\infty} \right]$$

$$= \frac{2\alpha}{\beta^2 - \alpha^2} \left(\frac{1}{|\alpha|} - \frac{1}{|\beta|} \right) \frac{\pi}{2} = \frac{\pi \alpha}{\beta^2 - \alpha^2} \left(\frac{1}{|\alpha|} - \frac{1}{|\beta|} \right) = \frac{\pi}{\beta^2 - \alpha^2} \left(\operatorname{sgn}(\alpha) - \frac{\alpha}{|\beta|} \right)$$

因为 $I(\alpha) = I(-\alpha)$, 所以只需讨论 $\alpha \geq 0$ 的情况

$$I(\alpha) = \int_0^{+\infty} \frac{\pi}{\beta^2 + t^2} \left(1 - \frac{t}{|\beta|} \right) \, dt = \pi \int_0^{+\infty} \left[\frac{1}{2|\beta|} \left(\frac{1}{|\beta|t} + \frac{1}{|\beta|t} \right) + \frac{1}{2|\beta|} \cdot \frac{t}{t^2 + \beta^2} \right] \, dt$$

$$= \pi \cdot \left[\frac{1}{2|\beta|} \left(\ln \left| \frac{t+|\beta|}{t-|\beta|} \right| \right) \Big|_0^\alpha + \frac{1}{2|\beta|} \ln(t^2 - \beta^2) \Big|_0^\alpha \right] = \frac{\pi}{2|\beta|} \ln(t+|\beta|)^2 \Big|_0^\alpha$$

$$= \frac{\pi}{|\beta|} \ln \left| \frac{\alpha+|\beta|}{1+|\beta|} \right|$$

$$\text{因此 } I(\alpha) = \begin{cases} \frac{\pi}{|\beta|} \ln \left(\frac{\alpha+|\beta|}{1+|\beta|} \right), & \alpha \geq 0 \\ \frac{\pi}{|\beta|} \ln \left(\frac{|\beta|-\alpha}{1+|\beta|} \right), & \alpha < 0 \end{cases} = \frac{\pi}{|\beta|} \ln \left(\frac{|\alpha|+|\beta|}{1+|\beta|} \right)$$